Cells & Development 168 (2021) 203720

Contents lists available at ScienceDirect

Cells & Development

journal homepage: www.journals.elsevier.com/cells-and-development

The cell in the age of the genomic revolution: Cell Regulatory Networks

Nicole Gorfinkiel [a], Alfonso Martinez Arias [b,*]

[a] Departamento de Genética, Fisiología y Microbiología, Facultad de CC Biológicas, Universidad Complutense, José Antonio Nováis 12, Madrid, Spain
[b] Systems Bioengineering, DCEXS, Universidad Pompeu Fabra, ICREA (Institució Catalana de Recerca i Estudis Avançats), Doctor Aiguader 88, Pg. Lluís Companys 23, Barcelona, Spain

ARTICLE INFO

Keywords:
Cells
Embryos
Genes
Gene Regulatory Networks
Cell Regulatory Networks

ABSTRACT

Over the last few years an intense activity in the areas of advanced microscopy and quantitative cell biology has put the focus on the morphogenetic events that shape embryos. The interest in these processes is taking place against the backdrop of genomic studies, particularly of global patterns of gene expression at the level of single cells, which cannot fully account for the way cells build tissues and organs. Here we discuss the need to integrate the activity of genes with that of cells and propose the need to develop a framework, based on cellular processes and cell interactions, that parallels that which has been created for gene activity in the form of Gene Regulatory Networks (GRNs). We begin to do this by suggesting elements for building Cell Regulatory Networks (CRNs). In the same manner that GRNs create schedules of gene expression that result in the emergence of cell fates over time, CRNs create tissues and organs i.e. space. We also suggest how GRNs and CRNs might interact in the building of embryos through feedback loops involving mechanics and tissue tectonics.

"*Despite the temptation to study morphogenesis, most developmental biologists have concentrated on the molecular mechanisms that control cell fate specification in embryos, and the enormous progress in understanding transcriptional control and intercellular signaling is obvious. However, morphogenesis has not been ignored.*"

Wallingford, Fraser and Harland (2002)

We live in an era dominated by the genetic analysis of biological systems. For example, the development of an organism is more often than not seen as a sequence of changes in gene expression driven by cocktails of transcription factor and signalling networks, a mapping of genes into cells. More recently, the arrival of single cell transcriptomics and epigenomics has added granularity to this view. However, the cell and not the gene, is the unit of development. This was first recognized in the second half of the XIX when, decades before an understanding of the nature and function of the gene, the observations and generalizations of Schwann, Schleiden and Virchow (Harris, 1999) were followed by a deluge of descriptive embryology at the cellular level. Embryos came to be seen as sequences of changes in cell shapes, numbers and higher level organization. Particularly important at this time are the lineage analysis of marine invertebrates by Conklin, Wilson and Chabry (Fischer, 1991; Wilson, 1925). These studies translated into the early days of experimental embryology in which, as Spemann apocryphally said: embryos were put in ever more embarrassing positions and asked to get out of them (Stern, 1954). Thus using sea urchin embryos, Driesch and Horstadius, revealed the regulative nature of early embryos and that, in many instances, fixed lineages hide conditional fates (Driesch, 1908; Horstadius, 1973). At the same time, work with amphibian embryos led Spemann and his school to show how these principles extended to vertebrate embryos, revealing a cellular framework of development with the gastrula organizer as the iconic piece of these studies (Spemann, 1962). However, by the middle of the century, this approach came to be stuck because a gap was growing between the phenomena that were being observed, the reproducible and effective responses to experimental intervention and their underlying causes (Hamburger, 1988). The missing piece was the gene and its function, particularly in development. Filling this gap had started earlier from what Sydney Brenner has called "Morgan's deviation" (Judson, 1996).

Thomas H Morgan was, by training and mindset, an embryologist who became distracted by trying to understand the material basis of phenotypic differences and developed Genetics on the way (Allen, 1968). Perhaps because he treaded both worlds, Genetics and Embryology, the need for a bridge between genetics and development was very prescient in his mind and he developed it in his Nobel lecture "The relation of Genetics to Physiology and Medicine" (Morgan, 1935). It is

* Corresponding author.
E-mail addresses: ngorfink@ucm.es (N. Gorfinkiel), alfonso.martineza@upf.edu (A. Martinez Arias).

https://doi.org/10.1016/j.cdev.2021.203720
Received 5 April 2021; Received in revised form 5 July 2021; Accepted 7 July 2021
Available online 9 July 2021
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Morgan who sets the path to understand the nature and function of genes and in doing so paves the way for the genetic analysis of development, which has been so successful. The mutant screens of the 1980s and 1990s (Anderson and Ingham, 2003) led to a gene centred explanation of embryonic development, a substitution of the cell for the gene. The cell becomes, simply, a by-product of the gene, a passive agent of its function, a capsule within which genes do their business. This change of emphasis is supported by the output of the first wave of mutant screens, largely in *C. elegans* and *Drosophila*, which identified a collection of conserved transcription factors and signalling molecules and leads to the notion of master regulators (Gehring and Ikeo, 1999). This research program has reached a climax recently when single cell RNAseq has provided granularity to the notion of "the cell" as a list of genes.

However, over the last few years, advances in the imaging of live cells and tissues together with the development of quantitative tools to analyse cell and tissue dynamics have resulted in a new view of embryonic development (Keller, 2013), shifting the focus from the regulation of gene expression back to the coordination of cell behaviours to generate form. This change has created a need to bridge a growing gap between the genetic and cell biological description of the development of an organism.

Here, as an introduction to this special issue of the new Cells and Development we want to comment on the present status of the relationship between genes and cells and the possibilities and likely insights gained from a refocusing on the cellular analysis of development in the age of genomics. We don't aim to be exhaustive but to pin point some significant examples of an emerging trend, many of these being discussed in further depth in this volume.

1. Genes and cells

If there is an event that highlights the role that cells play over genes in the building of an organism, it is gastrulation, particularly in vertebrate embryos (Stern, 2004). During gastrulation, a mass of seemingly equivalent cells engages into a carefully choreographed series of stereotypic cell movements that results into the outline of an organism. Traditionally, gastrulation is associated with the emergence of the germ layers. However, in many organisms e.g. *Drosophila*, echinoderms, amphibia and fish, these territories have been defined before the relative cell movements start, and in others e.g. mammals, the emergence of the germ layers is concomitant with the increase in embryo size and the acquisition of a coordinate system. Therefore, rather than as a stage associated with the emergence of the germ layers, a more accurate view of gastrulation is, perhaps, as a process whereby the embryo acquires a coordinate system and uses it to position the primordia of tissues and organs. The outcome of gastrulation is the phylotypic stage or process, at which all animals share a common body plan, which seems to act as a morphogenetic attractor (Duboule, 1994; Raff and Kaufman, 1983; Sander, 1976; Slack et al., 1993).

A transcriptional analysis of gastrulation returns a sequence of gene expression that appears to be conserved across vertebrates and, in some elements, in invertebrates (Abzhanov, 2013; Kalinka et al., 2010). Thus, transcription factors such as Brachyury and Snail and their regulation by BMP, Nodal and Wnt/β-catenin signalling are broadly conserved in the initial organization of the body plan. Nonetheless, it is clear that to understand gastrulation, one needs to go beyond the regulation of gene expression and consider the activities of cells. Individual cell behaviour is driven by the activity of the cytoskeleton, which is linked to adhesion systems that connect the cells to their neighbours and to the extracellular matrix. The coordinated activity of these modules of cellular activity generates different behaviours such as epiboly, invagination, Convergent and Extension (CE) and cavitation, that through the generation of local **stresses and strains** create the **volumes**, **geometries** and **boundary conditions** that shape the organism.

How gene expression and the mechanochemical activities of cells and tissues are connected during embryonic development was presciently discussed by Pere Alberch in an evolutionary context (Alberch, 1989) and is an important current topic of discussion (Collinet and Lecuit, 2021; Gilmour et al., 2017; Goehring and Grill, 2012; Gross et al., 2017). In *Drosophila*, the suite of pair rule transcription factors imposes polarity to the collective cell movements involved in Germ Band Extension (Irvine and Wieschaus, 1994; Lavalou et al., 2020; Paré et al., 2014; Zallen and Wieschaus, 2004) and in vertebrates, the transcriptional regulation of genes associated with the activity of the cytoskeleton and adhesion by transcription factors such as Bra and Snail (Barrallo-Gimeno and Nieto, 2005; Evans et al., 2012) points at an essential connection between these two levels of molecular organization. This is also emphasized by the outcome of mutant screens in mice which, though never systematically accounted for, does yield a large number of regulators of cell behaviour. This suggests that the connection between GRNs and cellular activity might be simply to incorporate genes coding for cytoskeletal and adhesion proteins and their regulators into GRNs. However, maps of gene expression tell us about cell identities and temporal sequences of GRN activity, but not, in principle, about how these result in form and much less about the dynamics of form. To do this, perhaps we need to consider a way to conceptualize cellular activities.

GRNs are a representation of gene expression; their nodes are Transcription Factors and, sometimes, signals and signalling pathway effectors. Their assembly can result in "circuits", small networks that perform specific operations usually associated with motifs, e.g. coherent Feed Forward Loops (FFL) that provide temporal control in gene switching and incoherent FFLs that can lead to oscillations and perform adaptation (Alon, 2019) These circuits define programs of gene expression underlying cell fate decisions. GRNs can be easily related to visions of development like the Waddington landscape and thus provide the elements for a temporal understanding of cell differentiation (Ferrell, 2012; Moris et al., 2016). The advent of single cell transcriptomics has extended this gene centric vision and allowed a detailed analysis of cell states and cell state transitions. However, GRNs do not contain and cannot provide spatial information: volumes, geometries, densities and the associated mechanochemical signalling. For example, GRNs of gastrulation can be interpreted and superimposed on the embryo but do not enlighten our understanding of the processes and cellular activities associated with the morphogenetic movements that shape an embryo through this process; they do not contain spatial information. This leads us to suggest the concept of Cell Regulatory Networks (CRNs), that follows some of the notions developed by Shook and Keller (2008) of 'morphogenic machines'. When building CRNs, it will be important to account for fundamental differences between, genes and cells. While GRNs are driven, and represented by interactions between genes (DNA) and transcription factors or signalling molecules (proteins), the core of CRNs are protein-protein interactions. Thus, the nodes of CRNs would be the components of cytoskeletal, adhesion and traffic related molecules, that could be represented in a 'gene code' but on the understanding that, operationally, their agents are proteins. Their interactions give rise to a basic toolbox of cell behaviours, i.e. cell division, growth, death, shape change and movement (Larsen and McLaughlin, 1987), that when coordinated at a multicellular level generate 3D tissue level processes such as epiboly, tissue invagination, CE, EMT, collective cell migration, tissue cavitation. These tissue level processes define volumes, shapes that can be seen as equivalent of the cell states in the GRNs and are dependent on the architectural context where they operate (see also Shook and Keller, 2008). Thus, GRNs are the engines of fates, identities, programs of gene expression, time in development. CRNs are the core of morphogenesis, mechanochemical signalling, and the creation of space in development (Fig. 1).

The notion that there might be two independent systems, chemical and mechanical, operating in parallel during the construction of an organism was first suggested by Turing on theoretical grounds to ease his analysis by focusing on the chemical component (Turing, 1952). Recently, work with Pluripotent Stem Cells (PSCs) has shown that this

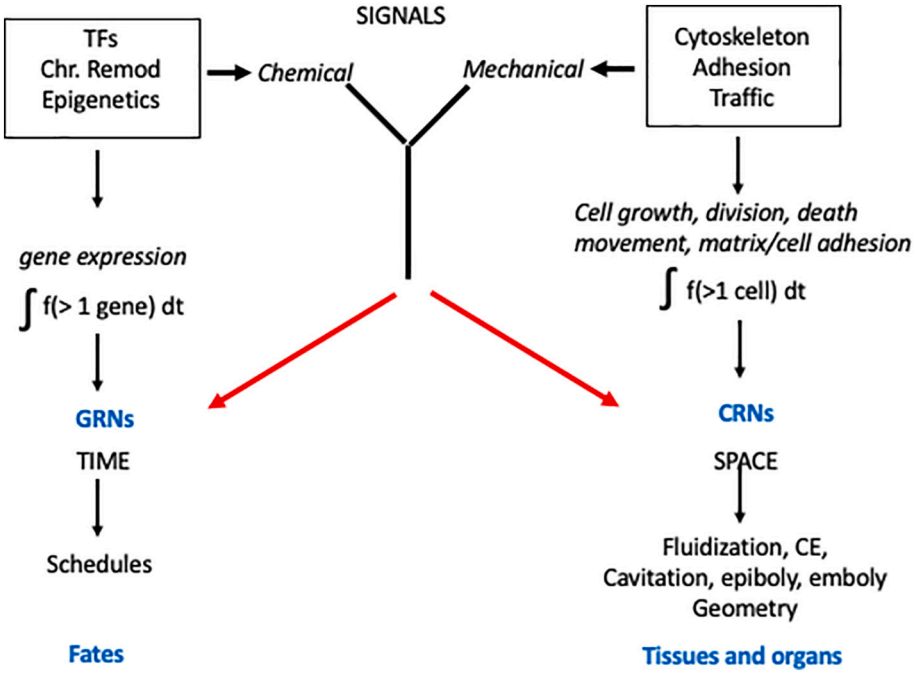


Fig. 1. Structural organization and comparison between Gene and Cell Regulatory Networks (GNR and CRN). See text for details.

separation might not just be a theoretical consideration but that a separation between chemical (gene based networks) and mechanical (cell based networks) might be real. Gastruloids, PSC models of gastrulation and body plan formation, exhibit a spatially proportionate organization of gene expression much as an embryo does but lack morphogenesis (Beccari et al., 2018; Moris et al., 2020a). However, morphogenesis can be induced by interactions with Matrigel, a natural matrix with multiple ECM components embedded in a complex mechanical environment (van den Brink et al., 2020; Veenvliet et al., 2020). This interaction probably triggers the activity of the CRNs.

With this conceptual outline in mind, we now explore the role that CRNs play in the morphogenesis of an embryo to then discuss the relationship between CRNs and GRNs.

2. Intrinsic cellular activities

Detailed descriptions of the behaviour of CRNs within whole tissues and embryos, from subcellular to supracellular/tissue scales, have been fuelled by major advances in microscopy, image analysis and quantitative cell biology over the last decades (Aigouy et al., 2010; Blanchard et al., 2009; Guirao et al., 2015; Keller, 2013). The integration of the activity of the cytoskeleton, cell-cell and cell-matrix adhesion and intracellular trafficking systems (the nodes of CRN) generate mechanical forces at the scale of single cells that, when integrated at the level of a multicellular systems, creates geometries and deformations that characterize tissues and organs. Moreover, these higher order behaviours can in turn impact on the activities of individual cells, CRNs as well as GRNs (Collinet and Lecuit, 2021; Gilmour et al., 2017; Heisenberg and Bellaiche, 2013; Petridou et al., 2017) through what is called 'downward causation', a feature that underlies many of the self-organizing properties of multicellular systems (Busby and Steventon, 2020; Collinet and Lecuit, 2021).

Well-studied examples of the activities of CRN where local and supra-cellular mechano-chemical feedback loops create space, geometry and deformations are tissue bending and tissue elongation. For example, during gastrulation, ventral epithelial cells undergo coordinated apical constriction involving the periodic contractions of the apical surface driven by a pulsatile actomyosin network (Martin et al., 2009). Cells eventually elongate in the apico-basal axis due to the basal displacement of the cytoplasm (Gelbart et al., 2012). Subsequent active shortening in the apico-basal axis driven by non-muscle Myosin II (Gracia et al., 2019) together with basal expansion (Krueger et al., 2018), are required to complete the invagination process. Tension-dependent intercellular interactions between cells are important for the correct invagination of mesodermal cells. Mesectodermal cells, i.e. the lateral cells of the prospective mesoderm, instead of contracting are stretched by their contracting neighbours in spite of having sufficient and functional actomyosin networks, and this is important to enable invagination (Bhide et al., 2020). Also, the observed anisotropy in the apical constriction of ventral epithelial cells results from the mechanical constraints imposed by the geometry of the whole tissue (Izquierdo et al., 2018; Martin et al., 2010). Apically constricting cells are mechanically coupled through a multicellular actomyosin network connected to adherens junctions (Martin et al., 2010) and a tension-dependent mechanism for the entrainment of apical constriction between neighbouring cells has been suggested (Xie and Martin, 2015).and collectively they generate tissue hydrodynamic viscous flows that ultimately produce the invagination of the tissue (He et al., 2014). Interestingly, global patterns of tissue hydrodynamic flows at the whole embryo level can be predicted to arise from the spatial distribution of Myosin II through local and non-local effects, the latter providing further evidence for mechanical coupling between cells (Dicko et al., 2017; Streichan et al., 2018).

Tissue elongation is another example that arises from the feedback between cell activity and mechanics. Germband extension in Drosophila mostly relies on cell intercalation, a multicellular process where the polarized shrinkage of cell interfaces in one direction and the formation and elongation of new interfaces in the orthogonal direction produces the extension of the embryonic epithelium (Bertet et al., 2004; Blankenship et al., 2006). Polarized junction shrinkage depends on the polarized flow of actomyosin pulses towards shrinking interfaces where Myosin II accumulates and induces anisotropic cortical tension (Rauzi et al., 2008; Rauzi et al., 2010) as well as on the sliding of tricellular vertices (Vanderleest et al., 2018).. When several aligned cells shrink their junctions in a coordinated manner they give rise to the formation of rosettes, multicellular structures that have been suggested to arise by the

tension-dependent recruitment of Myosin (Blankenship et al., 2006; Fernandez-Gonzalez et al., 2009). The formation of new junctions also depends on stresses generated by medial Myosin II pulses in cells surrounding the new junction and are oriented by tissue-scale pulling forces present at the posterior of the embryo (Collinet et al., 2015; Yu and Fernandez-Gonzalez, 2016).

Pulsatile actomyosin contractile forces arise from cell-intrinsic mechano-chemical feedback loops involving Rho1/A oscillations (Mason et al., 2016; Michaux et al., 2018; Munjal et al., 2015)and need to be stabilized to generate permanent deformations and thus prevent the reversible elastic recoil (Clément et al., 2017). Sources of stabilization may come from the endocytic machinery, by enabling cell membrane internalization, coupled to actomyosin contractility (Jewett et al., 2017) and from the turnover timescales of actin filaments and cross-linkers, which are typically shorter than the duration of contractile forces, implying that newly actin filaments are generated with no memory of previous stored elastic energy (Clément et al., 2017).

In contrast to what happens in *Drosophila*, where gastrulation and convergent and extension are distinct morphogenetic movements driven by apical constriction and cell intercalation respectively, vertebrate gastrulation involves a combination of cell ingression, cell division and cell intercalation. Cell ingression refers to the invagination of single cells, which undergo epithelial to mesenchymal transition involving apical constriction and elongation in the apico-basal axis (Voiculescu et al., 2014; Williams et al., 2012). In chick embryos, these cell shape changes are first observed in individual cells scattered across the tissue and then in an increasingly coordinated manner at the level of the primitive streak as gastrulation takes place (Rozbicki et al., 2015; Voiculescu et al., 2014). In mouse embryos, in situ epithelial to mesenchymal transitions take place in the posterior region of the epiblast and progresses anteriorly, and also involve apical constriction and translocation to the underlying mesodermal layer (Williams et al., 2012). Although the precise cellular mechanisms underlying cell ingression have not been described yet, it is likely that the actomyosin cytoskeleton is involved and it will be very interesting to investigate if tension-dependent mechanisms are involved in regulating these processes. In the chick embryo, cell intercalation involving the polarized shrinkage of junctions in the central region of the epiblast underlies the individuation of the primitive streak, its extension and thickening (Rozbicki et al., 2015; Voiculescu et al., 2007; Voiculescu et al., 2014). Supracellular Myosin II cables that are aligned in the direction of shrinkage are observed at cell junctions in this region and interfering with this localization partially impairs streak formation. This suggests the existence of Myosin II-dependent mechanisms, but also of other mechanisms, likely dependent on Myosin I and V, involved in junctional contraction. It has been suggested that the formation of these supracellular cables could be tension-dependent, as it has been observed in Drosophila (Rozbicki et al., 2015).

A novel cell intercalation behaviour induced by cell division has been described in the epiblast of the avian embryo anterior and lateral to the primitive streak. While in several epithelia embryonic epithelia after cell division, daughter cells remain in contact (Herszterg et al., 2013; Pinheiro et al., 2017), in the epiblast undergoing gastrulation, neighbouring cells to a mitotic cell intercalate in between daughter cells or a daughter cell itself intercalate between its neighbours (Firmino et al., 2016). These cell intercalation events induced by cell division require a relative depletion of actomyosin from the basolateral interfaces and a higher dynamic Myosin II apical pool in neighbouring cells to the mitotic cell (Firmino et al., 2016). Such cell intercalation events are likely to provide the basis for the fluid-like properties of the epiblast. Under the framework of a fluid-mechanical model, it has been proposed that the fluid-like properties of the epiblast are important for the propagation of forces generated by contractile actomyosin cables at the posterior margin between the embryo proper and the extraembryonic tissue. This suggests that local generated forces, through mechanical coupling between cells, can have an impact in the establishment of the body plan at the whole embryo level (Saadaoui et al., 2020).

Previously described mechanisms of cell intercalation involve the formation of bipolar mediolateral actin protrusions driven by the actomyosin cytoskeleton, which were mostly observed in mesodermal cells undergoing convergent and extension movements (reviewed in (Sutherland et al., 2020)). Although cell intercalation driven by mediolateral protrusions and the above described polarized junction shrinkage were initially considered to pertain to mesoderm and epithelial tissues respectively, recent observations suggest that these two mechanisms coexist (reviewed in Huebner and Wallingford, 2018). For example, in the Drosophila germband, basolateral protrusions have been shown to contribute to cell intercalation in rosettes (Sun et al., 2017). In the Xenopus mesodermal cells of the marginal zone, where mediolateral protrusions driven by a dynamic actomyosin cytoskeleton were initially observed (Shih and Keller, 1992) (Kim and Davidson, 2011; Skoglund et al., 2008), mediolateral junctions are enriched in phosphorylated Myosin II and exhibit enhanced tension, which drive polarized junction shrinkage similar to the Drosophila germband (Shindo and Wallingford, 2014). Both mediolateral basolateral protrusions and mediolateral apical junction shrinkage also act in concert during the convergence and extension of the mouse neural plate (Williams et al., 2014).

3. The coordination of cellular activities: Planar Cell Polarity (PCP)

In all embryos, a central feature of morphogenetic processes is the spatio-temporal coordination of the activities of individual cells. The processes associated with the emergence of the metazoan body plan during the extension along the anteroposterior axis are a good example of this. The intercalation of epithelial cells in the blastoderm of *Drosophila* (Lecuit and Lenne, 2007), the CE movements associated with gastrulation in *Xenopus* (Shook and Keller, 2003; Shook and Keller, 2004) and zebrafish (Gray et al., 2011; Yin et al., 2008) or axial extension in mammalian embryos (Sutherland et al., 2020) require a high degree of coordination of the activities of the participating cells. For example in *Xenopus* and zebrafish cells move mediolaterally which demands some form of molecular gyroscope and the synchronization across a population to ensure a smooth shape transition. Genetic analysis of these processes reveals the need for elements of the Planar Cell Polarity branch of Wnt signalling, as essential for these events. The Wnt-PCP pathway, as it is called, encompasses a number of membrane associated proteins and a subset of Wnt ligands that control the spatial orientation of cytoskeletal activity and organization (Butler and Wallingford, 2017). Embryos mutant for members of the Wnt-PCP pathway exhibit disorganized and uncoordinated cell movements and transitions that, over time, lead to severe disruptions of tissue organization. These mutants reappear in screens for elements involved in tissue and organ polarity and organization. These observations suggest that elements of the Wnt-PCP signalling pathway are the components of a cellular gyroscope with a higher level component that allows the coordination of these activities in space and time (Mateus et al., 2009). There might be other ways of implementing the intercellular coordination that is necessary for morphogenesis but Wnt-PCP signalling might be a central element of this operation, acting on CRNs in the same manner that it has been suggested that Wnt-β-catenin signalling acts in GRNs (Muñoz-Descalzo et al., 2012), which together with other signals, channels information to the nodes of CRNs, the cytoskeleton, adhesion and traffic systems.

4. Tissue-level properties: materials and mechanical interactions

Sculpting embryos and organisms from tissues also requires spatio-temporal variations in tissue mechanical properties such as fluidisation and stiffening, which are important in determining the extent and pace

of tissue deformation in response to active and passive forces (Davidson, 2011; Davidson et al., 2009). This demands a new level of description, the formulation of frameworks to represent and understand this and to provide a downward connection between them and CRN activities.

At the onset of gastrulation in zebrafish, the blastoderm begins to thin and the yolk underneath bulges upward in a process called doming. Recent studies have shown that the spatio-temporal fluidisation of the blastoderm underlies the thinning of the tissue. The viscosity of the central region of the blastoderm measured with micropipette aspiration rapidly decreases at this stage, which has been suggested to be important to decrease the resistance to the forces generated by the expansion of the covering enveloping layer. This tissue fluidisation results from an increase in the interstitial fluid between cells and thus a decrease in cell-cell contact number, size and duration (Petridou et al., 2019). Importantly, the topology of cell-cell contacts or cell connectivity is the key microscopic parameter that predicts macroscopic tissue material properties in a non-linear manner, suggesting the existence of a phase transition (Petridou and Heisenberg, 2019; Petridou et al., 2019; Petridou et al., 2021). Two sequential rounds of cell cleavage produce a decrease in the cell connectivity network and may provide a cellular mechanism to ensure the robust fluidisation of the blastoderm (Petridou et al., 2021).

The fluid-like properties of other tissues have also been associated with the rearrangement of cell-cell contacts. For example, during tissue epiboly in *Tribolium*, the extra-embryonic serosa shows spatial differences in the mechanical properties. While the dorsal serosa shows solid-like properties, the ventral part in which cells undergo cell rearrangements, exhibits a fluid-like behaviour, which is likely to be induced by a non-homogenous actomyosin cable at the ventral leading edge. This fluidisation would be important to release the stresses generated by the expansion of the serosa over the egg and maintain the epithelial integrity of the tissue (Jain et al., 2020). A similar process has been observed during epithelial wound healing, where the actomyosin cable that forms at the edge of the wound promotes similar cell intercalation and tissue fluidisation (Tetley et al., 2019).

Conversely, fluid-to-solid transitions have also been observed during morphogenesis. During the elongation of the zebrafish embryo, as progenitor cells from the mesodermal progenitor zone differentiate and are incorporated in the more anterior presomitic mesoderm, the tissue undergoes a fluid-to-solid jamming transition. Using ferrofluid magnetic droplets, it has been found that the more anterior presomitic mesoderm is more rigid than the posterior mesoderm, which at the cellular level is associated with higher cell cohesion while the more fluid-like posterior mesoderm shows higher rates of cell intercalation (Mongera et al., 2018). It has been suggested that the rigidification of the anterior mesoderm is important to support the posterior elongation of the fluid-like posterior mesoderm (Mongera et al., 2018). Tissue stiffening is also present in the head mesoderm in Xenopus, but in this case this is linked to an increase in cell density resulting from convergent and extension movements of mesodermal cells (Barriga et al., 2018). Importantly, this stiffening is required and is sufficient to promote an epithelial to mesenchymal transition in overlying neural crest cells and trigger their collective migration using the head mesoderm as a substrate (Barriga et al., 2018).

Mechanical interactions between tissues had been uncovered during Drosophila germband extension, where the invagination of the ventral mesoderm as well as the invagination of the posterior endoderm actively contribute to the activities of cells driving the extension of the germband (Butler et al., 2009; Collinet et al., 2015; Lye et al., 2015). More recently, mechanical coupling between tissues has been shown to be important to coordinate the elongation of tissues from different germ layers in the chick embryo. Using a combination of elegant techniques to apply extrinsic forces and to prevent biochemical communication between axial (notochord, neural tube and progenitor domain) and paraxial tissues (presomitic mesoderm-PSM), a positive inter-tissue mechanical feedback loop between these tissues has been revealed (Xiong et al., 2020). The posterior PSM compresses the axial structures, contributing to the mediolateral convergence and anteroposterior extension of the notochord and neural tube. The extension of these axial tissues then pushes the progenitor domain posteriorly, which drives the lateral movements of progenitor cells into the posterior PSM, eventually contributing to the compression of the axial structures and closing the positive feedback loop (Bénazéraf et al., 2017; Xiong et al., 2020).

Finally, boundary conditions imposed by the geometry of the embryo and the extra-embryonic tissues, which are attached to the embryo, also provide inputs for the proper generation of body plans. In the beetle *Tribolium*, the integrin-mediated attachment of the anterior blastoderm to the vitelline membrane counteracts tissue-intrinsic contractile forces to generate the unidirectional posterior flow of the extra-embryonic serosa that eventually engulfs the embryo (Münster et al., 2019). Attachment of a developing tissue to an external structure has also been observed in the developing *Drosophila* wing, where the attachment of the epidermis to the external cuticle creates a resistive force key for the proper wing shape (Etournay et al., 2015; Ray et al., 2015). Such interactions are also important during avian gastrulation where the supracellular actomyosin cables at the interface between the epiblast and the extra-embryonic tissues create tensile forces that shape the embryo (Saadaoui et al., 2020). Also, changes in the mechanical properties of the vitelline membrane of the avian egg have an impact in the early expansion of the epiblast and the convergence of the body axis. These changes in the mechanical properties result from biochemical modifications of the glycoprotein layer of vitelline membrane due to an increase in the pH mediated by the release of CO_2 from the embryo (Kunz et al., 2021), revealing an unprecedented level of physiological regulation of development.

These results show the existence of mechanical interactions at the level of tissues, which are important not only for the mechanical coordination of whole embryo morphogenesis but are also likely to contribute in providing signals for the generation of the embryo body plan and thus provide the basis for downward causation.

5. Interactions between GRNs and CRNs: an integrated view of morphogenesis

As posed by Turing, the development of an organism is the result of an interaction between its chemical and mechanical components (Turing, 1952), between GRNs and CRNs, as suggested here. There is ample evidence that GRNs influence the activity of CRNs by changing the activity of some of their components. For example, as stated above, in Drosophila the transcriptional networks determining gastrulation and, also Germ Band Extension and dorsal closure, control the localization of cytoskeletal and Planar Cell Polarity regulators and thus set up the ground for large scale cell movements. This is probably also true in vertebrates where Brachyury, that drives gastrulation, regulates the expression of cytoskeletal and adhesion molecules with its main function being the localized regulation of directional movements. These examples show that transcription can exert some level of control over the activities of cells. However, the coordination of these activities at the tissue level cannot be a direct consequence of this regulatory relationship.

Morphogenesis cannot be understood as a linear process from genotype to phenotype but results from dynamic interactions between patterning signals, Cell Regulatory Networks and the final architecture of the embryo. An integration of these elements into a framework or the relationship between genes and cells has been proposed by Shook and Keller (2008) (Fig. 2) and suggested to account for how conserved patterning pathways lead to the phylotypic stage through a great diversity of developmental mechanisms. In this context, it is worth mentioning that aggregates of gastrula embryonic cells of the sea anemone *Nematostella vectensis*, a cnidarian model organism, re-establish the normal body plan but use different Cell Regulatory Networks than the normal organism (instead of forming the three germ

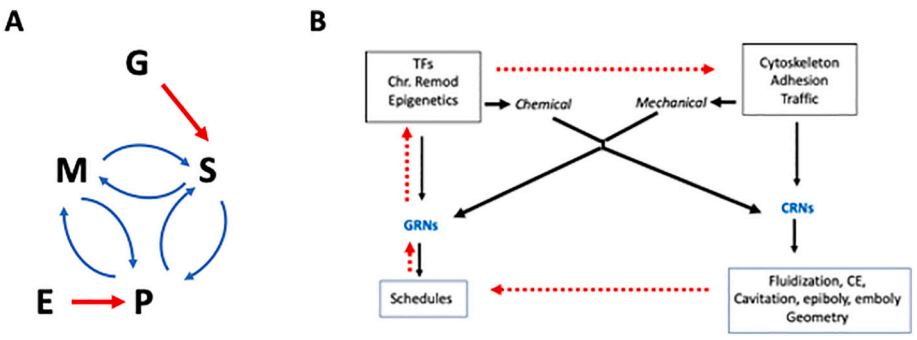


Fig. 2. Interactions between Gene and Cell Regulatory Networks. (A) Relationship between genes and morphogenesis as envision by Shook and Keller (2008). Genes (G for genotype in their account), control patterning signals (S) that direct 'morphogenic machines' (M) that create architectural phenotypes (P). M, S and P are engaged into a series of reciprocal interactions that change over time and regulate and coordinate the building of tissues and organs. The external environment (E) affects the morphogenetic processes through its effects on P. (B) In the vision of the chemical and mechanical components of development as two independent entities, their interaction has two levels. One is through the effects of the convergence of the chemical and mechanical signals on the structure and activity of GRNs and CRNs; this would be the equivalence of the effects of S on P and M in the vision of Shook and Keller. However, most importantly, the effect of E on these crucial elements is mediated by downward causation (red dotted arrows) whereby the mechanochemical properties of tissues and organs affect the activity and structure of GRNs, creating a feedback on the activities of cells (see text for further details). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

layers through invagination they use delamination, multipolar ingression and cavitation) (Kirillova et al., 2018), revealing the context dependent activity of Cell Regulatory Networks. This might also be the case with gastruloids, which reveal that developmental GRNs can be uncoupled from CRNs (Moris et al., 2020b). However, gastruloids suggest that, rather than being linked in a linear manner, the chemical and mechanical components of a developmental system are two independent entities brought together by 'weak linkage' by a selective process in which the phylotypic stage might act as a morphogenetic attractor (Busby and Steventon, 2020) (Fig. 2).

A role for supracellular mechanics and boundary conditions in the regulation of gene expression during gastrulation has been uncovered in different organisms. The mechanical tension generated by apically constricting cells in Drosophila triggers a mechanosensitive pathway involving the phosphorylation of Armadillo/β-catenin at tyrosine-667, which is released from cell junctions and enters the nucleus to maintain the expression of Twist, the Drosophila transcription factor for mesoderm identity (Brunet et al., 2013; Pouille et al., 2009). Such mechanosensitive mechanism of induction of mesodermal cell fate is conserved in zebrafish, where *no tail*, the orthologue of *Brachyury*, is induced at the marginal zone as a consequence of epiboly movements, also through the nuclear translocation of β-catenin (Brunet et al., 2013). More recently, the same β-catenin-dependent mechanotransduction mechanism for the activation of *Brachyury* expression has been observed in the sea anemone *Nematostella vectensis* (Pukhlyakova et al., 2018). Overall, these observations have led to the suggestion that the mechanical strains generated by gastrulation movements trigger a conserved mechano-sensitive pathway for mesoderm induction (Brunet et al., 2013). Interestingly, using human embryonic stem cells (hESC) cultured on geometrically defined patterned substrates such as squares and triangles has shown that at sites of maximal tension, i.e. corners, as measured by traction force microscopy, gastrulation-like nodes appear upon induction with BMP4. The presence of such nodes is revealed by the activation of the expression of a Brachyury reporter and relies on the phosphorylation dependent release of β-catenin from junctions and its subsequent translocation to the nucleus to activate transcription (Muncie et al., 2020).These observations suggest a role for Wnt/β-catenin signalling in mechanotransduction that might be general.

The Hippo pathway is the other important pathway that provides a fundamental link between gene expression and cell mechanics, tissue geometry and boundary conditions in many developmental events. Changes in the stiffness of the extracellular matrix and in the organization and activity of the actomyosin cytoskeleton have been shown to regulate the nuclear import and thus the activity of YAP/TAZ, the transcriptional coactivator of the pathway in different cultured cell systems (Totaro et al., 2018). Also, during preimplantation mouse development, differential contractility between blastomeres drives their sorting into inner and outer positions of the blastocyst, which defines the trophoectoderm and Inner Cell Mass lineages through position-dependent Hippo signalling. Interestingly, contractility also regulates YAP subcellular localization, suggesting that the Hippo pathway translates surface tension and tissue geometry into differential gene expression in these early developmental stages (Maître et al., 2016; Posfai et al., 2017). Moreover, during the development of the zebrafish hindbrain, the formation of actomyosin cables at the boundary between the different rhombomeres provides mechanical signals sensed by YAP/TAZ to maintain the proliferative state of boundary cells (Voltes et al., 2019). The mechanosensitivity of this pathway also plays an important role during tissue homeostasis as shown by the finding that the quiescent vs proliferative state of the muscle stem cell niche in adult mice is controlled by a Wnt-4 dependent mechanosensitive pathway that regulates YAP nuclear localization (Eliazer et al., 2019).

We suggest that a combination of Hippo and Wnt/β-catenin signalling provides an input from the tissue level properties of multicellular ensembles into the circuits that regulate cell fate and thus alter their operation in a manner that adjusts and alters the elements at the cell level (Fig. 2). These feedbacks create an integrated system with the capacity to integrate in a homeostatic manner the activities of GRNs and CRNs.

6. Concluding remarks

During the sustained, and perhaps necessary, period of focus on gene expression that completes Morgan's deviation questions about morphogenesis have not been ignored. The recovery of a view of the development of embryos as the result of cell interactions and the recognition that a gene centric view is not able to explain cellular dynamics or, at a higher level, morphogenesis, are now being pursued through the development of experimental and technical advances that will provide new questions and conceptual frameworks. This cell based view of development in the modern context requires a rich interaction between the physical and biological sciences which has been always at the core of cell biology but which now is extending to developmental cell biology and is represented by the mechanico-statistical analysis of the behaviour of cell populations (Aigouy et al., 2010; Blanchard et al., 2009; Guirao et al., 2015; Petridou et al., 2021), fluid mechanic approaches (Dicko et al., 2017; Saadaoui et al., 2020; Streichan et al., 2018) and notions such as tissue tectonics and downward causation (Busby and Steventon, 2020).

Notwithstanding this progress, the real challenge is, and will be, to find the structural and functional basis of the interactions between genes, cells and tissues which, as recognize early on by Turing adds

enormous difficulty to understanding the emergence of shape and form but which are central to the understanding of the development of an organism. The XX century developed as the century of the gene; it opened up with the rediscovery of Mendel's laws and ended with the unveiling of the human genome but the XXI century is bound to be the century of the cell and the change of name from Mechanisms of Development to Cells and Development and this inaugural issue bear witness to this.

Acknowledgements

We want to thank Roberto Mayor, for the invitation to submit this essay and to Ben Steventon for multiple discussions on the need to conceptualize the question of the relationship between chemical and mechanical signals in development. AMA research is funded by an ERC AdG (834580). NG is funded by grant PID2020-114533GB-C22 from the Ministerio de Ciencia e Innovación.

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